

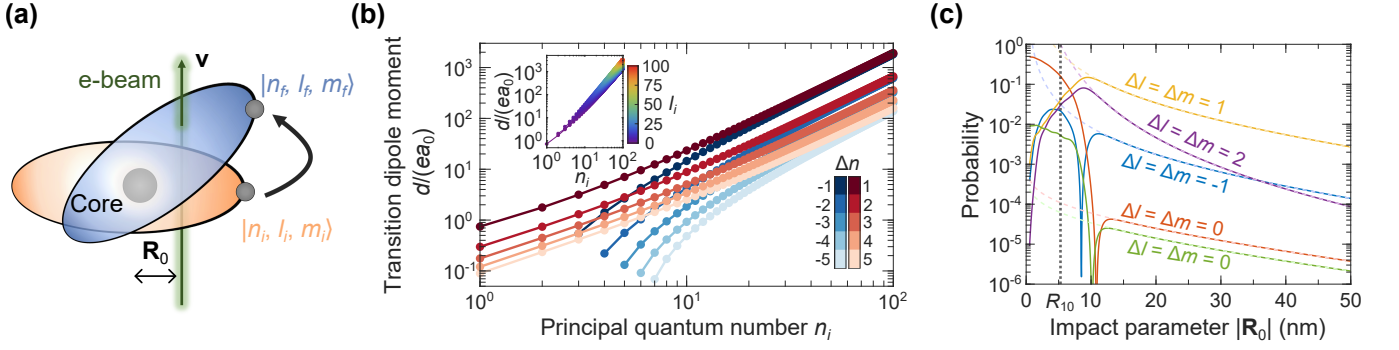


FIG. 1. **Interaction of Rydberg atoms with electron beams.** (a) The interaction of a Rydberg atom with a free electron moving along a straight line and passing at a position $\mathbf{R}_0$ from the nucleus (impact parameter) can drive transitions between initial and final states, $|n_i, l_i, m_i\rangle$ and $|n_f, l_f, m_f\rangle$, respectively. (b) Transition dipole moment $d = e\langle n_i, 0, 0 | \mathbf{r} | n_f, 1, 1 \rangle$ (in units of ea_0) associated with dipolar transitions in the Rydberg atom as a function of initial principal quantum number n_i for different values of $\Delta n = n_f - n_i$ (see color legend). The inset shows results for $|n_i, l_i, 0\rangle \rightarrow |n_i + 1, l_i + 1, 1\rangle$ transitions with a wide range of l_i values. (c) Impact-parameter dependence of the probability of $|10, 5, 5\rangle \rightarrow |11, 5 + \Delta l, 5 + \Delta m\rangle$ transitions for different values of Δl and Δm as produced by interaction with a 10 eV electron. First-order matrix-element calculations (solid curves) are compared to the large- $|\mathbf{R}_0|$ scaling $\propto K_{|\Delta m|}^2 (|\varepsilon_{fi}| R_0 / v)$ with $\hbar\varepsilon_{fi} = (1/n_i^2 - 1/n_f^2)$ Ry (dashed curves). A vertical broken line indicates the atomic radius $R_n = n^2 a_0$ with $n = 10$ and a_0 the Bohr radius.

that acts on the Rydberg atom, and we study their quantum-mechanical interaction by adopting the minimal coupling prescription [52] and neglecting A^2 terms. Then, the Hamiltonian of the system reduces to $\hat{\mathcal{H}}(\mathbf{r}, t) = \hat{\mathcal{H}}_0(\mathbf{r}) + \hat{\mathcal{H}}_1(\mathbf{r}, t)$, where $\hat{\mathcal{H}}_0(\mathbf{r}) = -(\hbar^2/2m_e)\nabla^2 + V(\mathbf{r})$ describes the Rydberg electron in a Coulomb-like core potential $V(\mathbf{r})$ while $\hat{\mathcal{H}}_1(\mathbf{r}, t) = -(ie\hbar/2m_e c)[\nabla \cdot \mathbf{A}(\mathbf{r}, t) + \mathbf{A}(\mathbf{r}, t) \cdot \nabla] - e\phi(\mathbf{r}, t)$ accounts for the perturbation produced by the swift electron through the vector and scalar potentials $\mathbf{A}$ and ϕ . Writing the electron trajectory as $\mathbf{r}_0(t) = \mathbf{R}_0 + \mathbf{v}t$ (with $\mathbf{R}_0 \perp \mathbf{v}$) and its associated electric field as $\mathbf{E}(\mathbf{r}, t) = -(1/c)\partial_t \mathbf{A}(\mathbf{r}, t) - \nabla\phi(\mathbf{r}, t)$, we have the potentials [53]

$$\phi(\mathbf{r}, t) = -\frac{e\gamma}{\sqrt{|\mathbf{R} - \mathbf{R}_0|^2 + \gamma^2(\mathbf{r} \cdot \hat{\mathbf{v}} - vt)^2}},$$

and $\mathbf{A}(\mathbf{r}, t) = (\mathbf{v}/c)\phi(\mathbf{r}, t)$, where $\gamma = (1 - v^2/c^2)^{-1/2}$ is the Lorentz factor. In what follows, we consider low-energy electrons ($v \ll c$), so we approximate $\gamma \approx 1$ and neglect the effect of the vector potential.

In general, the wave functions of the Rydberg states can be calculated analogously to those of the hydrogen atom using the so-called quantum defect theory [54], where the differences between the Coulomb potential of the proton in hydrogen and the actual potential experienced by a Rydberg electron are conveniently encapsulated in a species-dependent model potential with empirical parameters [1, 54, 55]. For simplicity, in this work, we consider Rydberg states described by hydrogenic wave functions $\psi_i(\mathbf{r})$ and neglect fine and hyperfine structure effects [56], so they satisfy $\hat{\mathcal{H}}_0(\mathbf{r})\psi_i(\mathbf{r}, t) = \hbar\varepsilon_i\psi_i(\mathbf{r}, t)$ with $V(r) = -e^2/r$ and eigenenergies $\hbar\varepsilon_i = (-1/n_i^2)$ Ry expressed in terms of the Bohr radius a_0 . Here, $i \equiv (n_i, l_i, m_i)$ encapsulates the principal, angular, and azimuthal quantum numbers. Hydrogenic states form a complete orthonormal basis set, and therefore, the time-dependent wave function can be expanded as $\psi(\mathbf{r}, t) = \sum_i \alpha_i(t)\psi_i(\mathbf{r})e^{-i\varepsilon_i t}$, where the expansion coefficients

$\alpha_i(t)$ satisfy the equation of motion

$$\dot{\alpha}_i(t) = -\frac{i}{\hbar} \sum_{j \neq i} e^{i\varepsilon_{ij}t} M_{ij}(t) \alpha_j(t) \quad (1)$$

with energy differences $\varepsilon_{ij} = \varepsilon_i - \varepsilon_j$ and $M_{ij}(t)$ and transition matrix elements defined by $M_{ij}(t) = \int d^3\mathbf{r} \psi_j^*(\mathbf{r}) \hat{\mathcal{H}}_1(\mathbf{r}, t) \psi_i(\mathbf{r})$ (Fig. 1a). [See Appendix A for details on the solution of Eq. (1).] This formalism can be straightforwardly applied to Rydberg electron wave functions resulting from more complex phenomenological core potentials $V(r)$ [1, 54, 57].

B. Interaction of a free electron with a Rydberg atom

Before solving the full dynamics described by Eq. (1), it is instructive to examine the interaction within first-order perturbation theory, where the transition probability becomes

$$P_{i \rightarrow f}^{(1)} = \frac{1}{\hbar^2} \left| \int_{-\infty}^{\infty} dt e^{i\varepsilon_{fi}t} M_{fi}(t) \right|^2 = \left(\frac{2e^2}{\hbar v} \right)^2 \left| \int_{-\infty}^{\infty} d^3\mathbf{r} \psi_f^*(\mathbf{r}) \psi_i(\mathbf{r}) e^{i\varepsilon_{fi}\mathbf{r} \cdot \mathbf{v} / v^2} K_0 \left(\frac{|\varepsilon_{fi}| |\mathbf{R} - \mathbf{R}_0|}{v} \right) \right|^2.$$

The electric dipole moment of a highly excited Rydberg atom exhibits a $\propto n^2$ scaling with the principal quantum number, reaching large values for $n \gg 1$ (Fig. 1b). In the dipolar limit (i.e., for e-beams passing far away from the atom, so that the interaction is dominated by dipolar transitions), the probability reduces to [33, 34]

$$P_{i \rightarrow f}^{(1)} = \left(\frac{2e\varepsilon_{fi}}{\hbar v^2} \right)^2 \left[|d_{fi,x}|^2 K_1^2 \left(\frac{|\varepsilon_{fi}| b}{v} \right) + |d_{fi,z}|^2 K_0^2 \left(\frac{|\varepsilon_{fi}| b}{v} \right) \right], \quad (2)$$

where $d_{fi,x}$ and $d_{fi,z}$ denote Cartesian components of the transition dipole moment $\mathbf{d}_{fi} = -e \int d^3\mathbf{r} \psi_f^*(\mathbf{r}) \mathbf{r} \psi_i(\mathbf{r})$, and we take the electron trajectory as $\mathbf{r}_0(t) = b \hat{\mathbf{x}} + vt \hat{\mathbf{z}}$. For small arguments